

Geometric Frustration in Heisenberg Antiferromagnet

- The Shastry-Sutherland Model (SSM) is given by an anisotropic spin-1/2 Heisenberg Hamiltonian on a lattice:

$$H_{SSM} = J' \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J \sum_{\langle\langle i,j \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j,$$

with nearest neighbor and next nearest neighbor interactions. J' and J are positive. Initially, it was a "Toy Model" for frustration in a 2D-lattice.

- The ground state of H_{SSM} is exactly known to be the product of *orthogonal* dimer singlets

$$|\psi_{dimer}\rangle = \prod |\mathcal{S}_{\langle i,j \rangle}\rangle$$

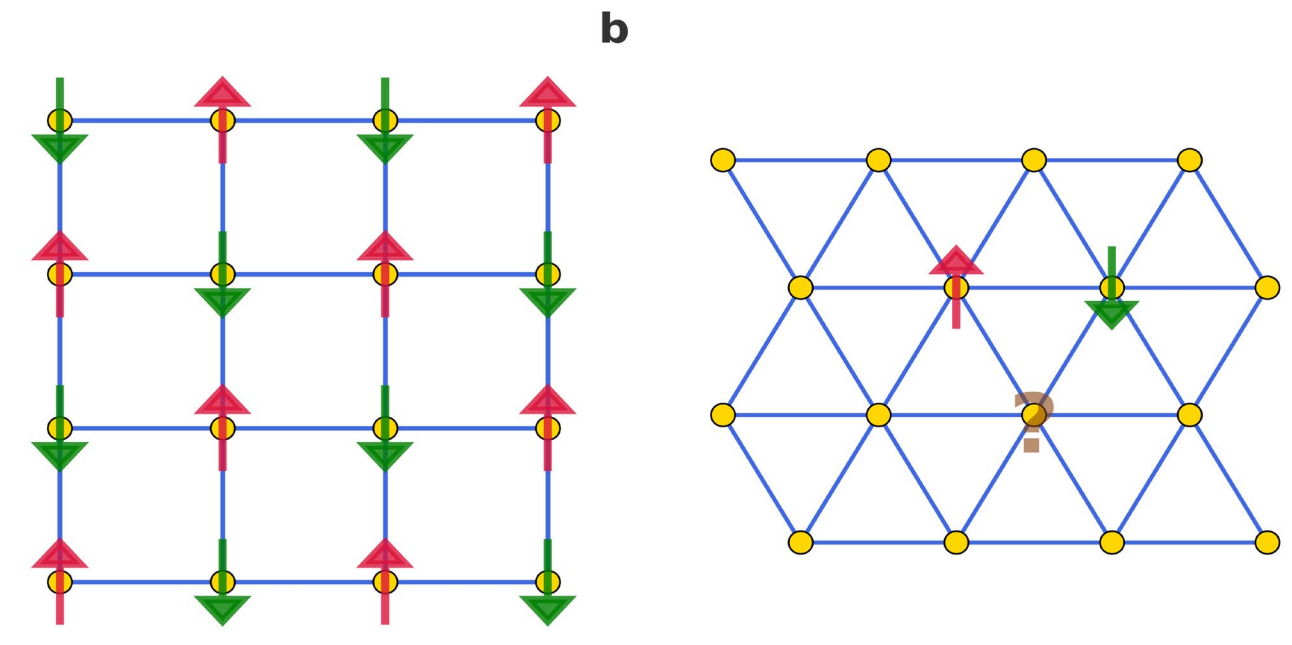
with $|\mathcal{S}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ for $J'/J \leq 0.5$.

- In 1999, H_{SSM} was discovered to be applicable in describing the physics of $\text{SrCu}_2(\text{BO}_3)_2$. Since then, series of materials have been confirmed to be described by the same model.

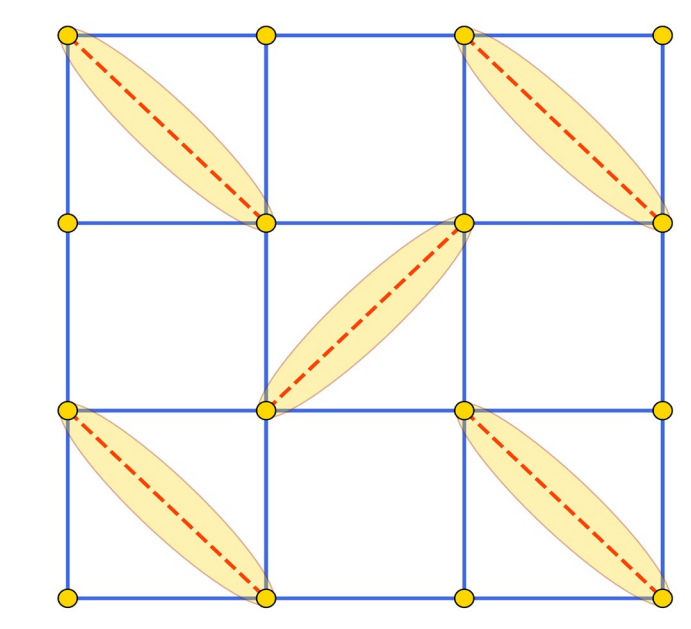
- Experimental evidence of a possible intermediate phase around $J'/J \approx 0.7$ between dimerized and Neel ordered state. The nature and existence of this phase continues to be debated.

- The model is analytically unsolved in the $J'/J > 0.5$ regime. Numerical methods have been applied in this sector. However, no complete understanding has yet been achieved.

- We develop an Exact Diagonalization (ED) and a Quantum Monte Carlo (QMC) algorithm to compute the ground-state energy of H_{SSM} .



Geometric Frustration



The Shastry-Sutherland Lattice

Exact Diagonalization and Quantum Monte Carlo

ED Program

$$H|\psi\rangle = E|\psi\rangle$$

- $\dim(H_{SSM}) = 2^{16}$ for a 16-site 4x4 lattice.

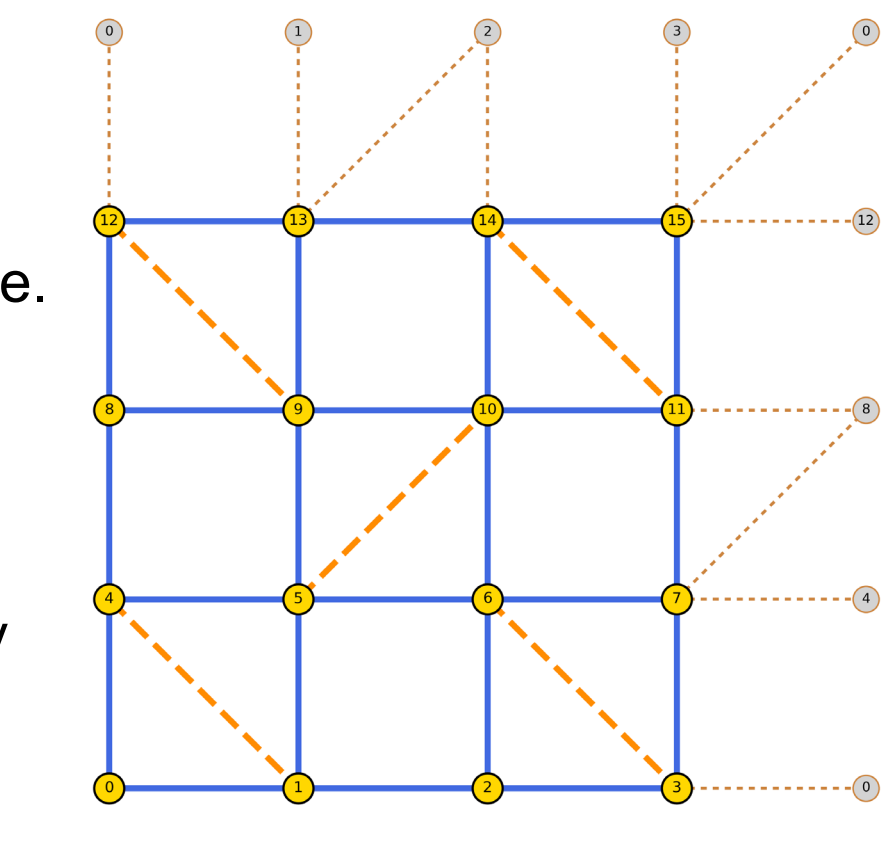
- Binary representation of eigenkets: $|s_1 s_2 \dots s_{16}\rangle$ with $s_i \in \{0,1\}$.

- In the $S_z = 0$ sector, the dimensionality reduces greatly:

$$\dim(H_{SSM})_{S_z=0} = \frac{16!}{8!8!} = 12,870.$$

- Periodic Boundary Conditions.

- Since $[H_{SSM}, \mathbf{S} \cdot \mathbf{S}] = 0$, where $\mathbf{S} = \sum_i \mathbf{S}_i$, we can obtain total spin of each energy eigenstate.



$$\begin{aligned} \mathbf{S}_i \cdot \mathbf{S}_j &= S_i^z S_j^z + \frac{1}{2} (S_i^+ S_j^- + S_i^- S_j^+) \\ \mathbf{S}_i \cdot \mathbf{S}_j |\uparrow\uparrow\rangle &= \frac{1}{4} |\uparrow\uparrow\rangle, \\ \mathbf{S}_i \cdot \mathbf{S}_j |\downarrow\downarrow\rangle &= \frac{1}{4} |\downarrow\downarrow\rangle, \\ \mathbf{S}_i \cdot \mathbf{S}_j |\uparrow\downarrow\rangle &= -\frac{1}{4} |\uparrow\downarrow\rangle + \frac{1}{2} |\downarrow\uparrow\rangle, \\ \mathbf{S}_i \cdot \mathbf{S}_j |\downarrow\uparrow\rangle &= -\frac{1}{4} |\downarrow\uparrow\rangle + \frac{1}{2} |\uparrow\downarrow\rangle. \end{aligned}$$

QMC Program

$$\begin{aligned} Z_{ij} &= \sum_{\psi_j} \langle \psi_i | e^{-\beta H} | \psi_j \rangle = \lim_{\epsilon \rightarrow 0, L_t \rightarrow \infty} \sum_{\psi_j} \langle \psi_i | (e^{-\epsilon H})^{L_t} | \psi_j \rangle \\ &= \sum_{\psi_j} \langle \psi_i | (1 - \epsilon H)^{L_t} | \psi_j \rangle = \sum_C W(C) S(C) \\ E_{ij} &= \sum_{\psi_j} \langle \psi_i | H e^{-\beta H} | \psi_j \rangle = \lim_{\epsilon \rightarrow 0, L_t \rightarrow \infty} \sum_{\psi_j} \langle \psi_i | H (e^{-\epsilon H})^{L_t} | \psi_j \rangle \\ &= \sum_{\psi_j} \langle \psi_i | H (1 - \epsilon H)^{L_t} | \psi_j \rangle = \sum_C W(C) S(C) E(C). \end{aligned}$$

Start Loop Update

Choose space-time site

Identify Closed Loop or Open String Connected to the Site

Flip all spins on closed loop or open string

Finish

Principle of Detailed Balance

$$P(C \rightarrow C') W(C) = P(C' \rightarrow C) W(C')$$

Ergodicity

The configuration space must be connected via the transition probabilities $P(C \rightarrow C')$.

Two kinds of Monte Carlo Updates;

- (i) Bond update: addition and removal of bonds
- (ii) Loop/Spin update: flipping of spins in a cluster

Start Bond Update

Toss a coin and check for heads

True

Propose to add a bond at a random space-time point

False

Propose to remove an existing bond at a spatial site

Accept with appropriate probability or reject

Finish

As $\beta \rightarrow \infty$, the matrix EZ^{-1} gives the low-lying spectrum of H .

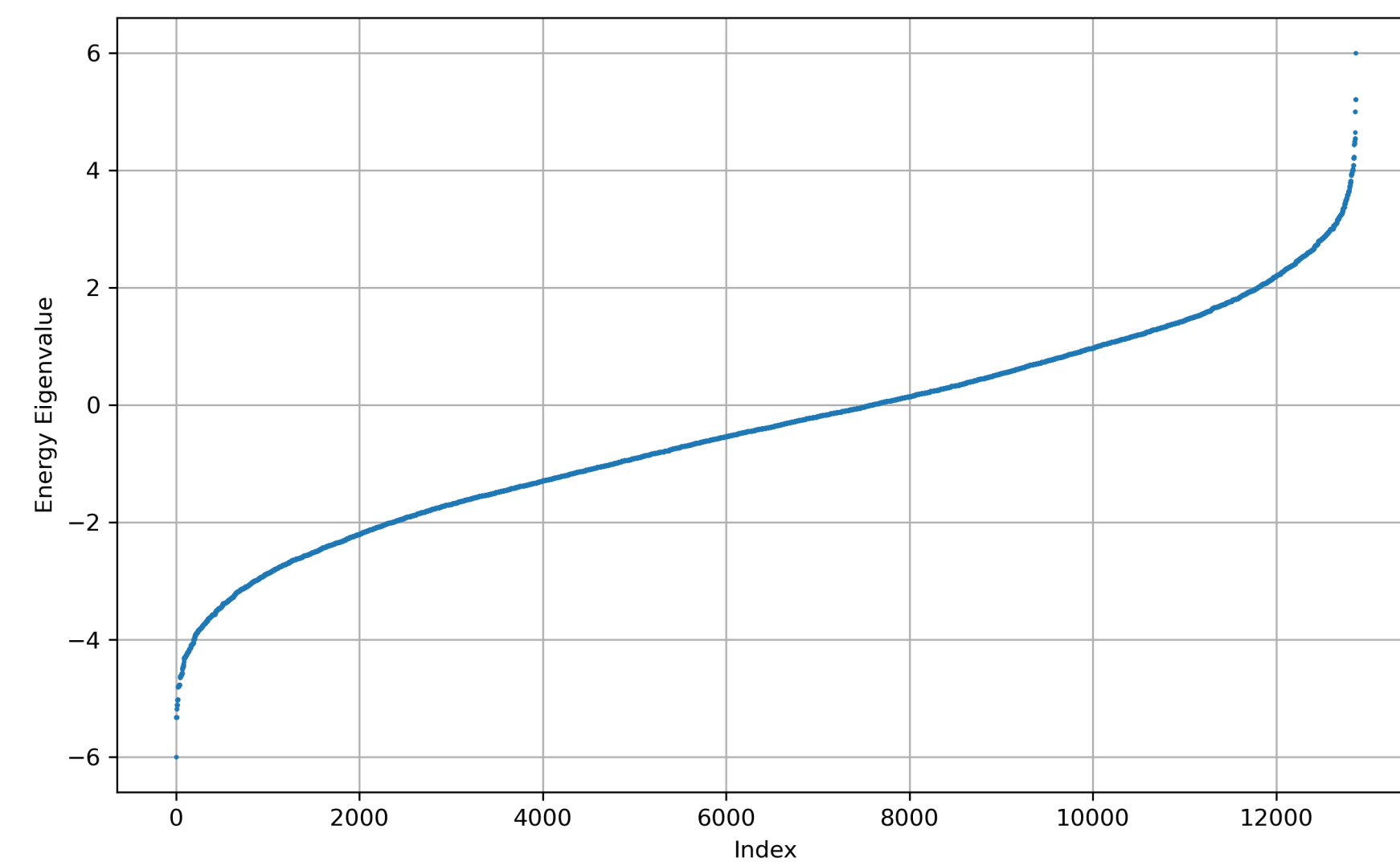
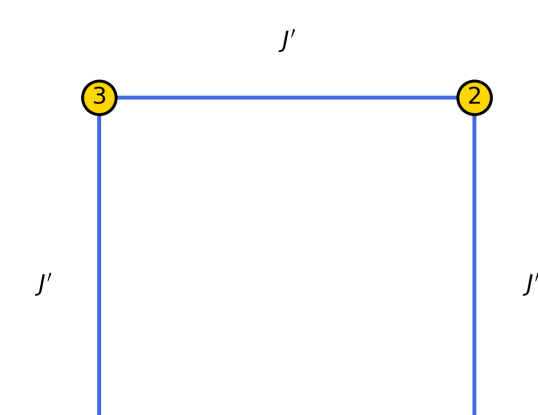


Figure: Sorted list of energy eigenvalues of the H_{SSM} in the $S_z = 0$ sector for $J=1$ and $J'=0.5$ a 16-site system.



$$H_{nn} = \mathbf{S}_0 \cdot \mathbf{S}_1 + \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_2 \cdot \mathbf{S}_3 + \mathbf{S}_3 \cdot \mathbf{S}_0 = (\mathbf{S}_0 + \mathbf{S}_2) \cdot (\mathbf{S}_1 + \mathbf{S}_3)$$

$$|S_{02}\rangle = \frac{1}{\sqrt{2}} (|\uparrow_0 \uparrow_2\rangle - |\downarrow_0 \uparrow_2\rangle), \quad (\mathbf{S}_0 + \mathbf{S}_2) |S_{02}\rangle = 0$$

$$H_{nn} |S_{02}\rangle = (\mathbf{S}_0 + \mathbf{S}_2) \cdot (\mathbf{S}_1 + \mathbf{S}_3) |S_{02}\rangle = 0$$

$$H_{SSM} |\psi_{dimer}\rangle = H_{nn} |\psi_{dimer}\rangle + H_{nnn} |\psi_{dimer}\rangle = -6J |\psi_{dimer}\rangle$$

Proof: The derivation showing that the dimer-product state is always the eigenstate of H_{SSM} .

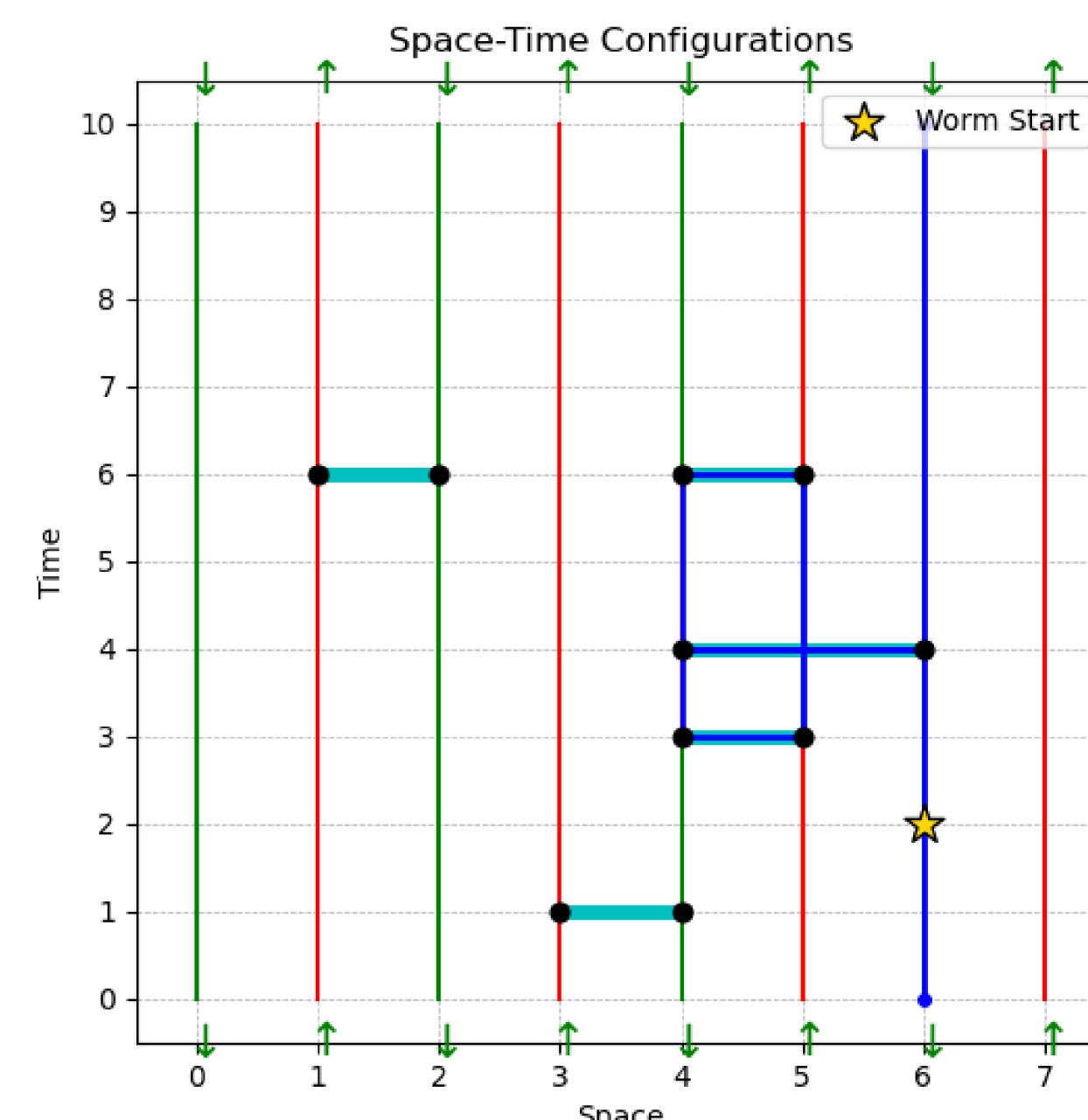
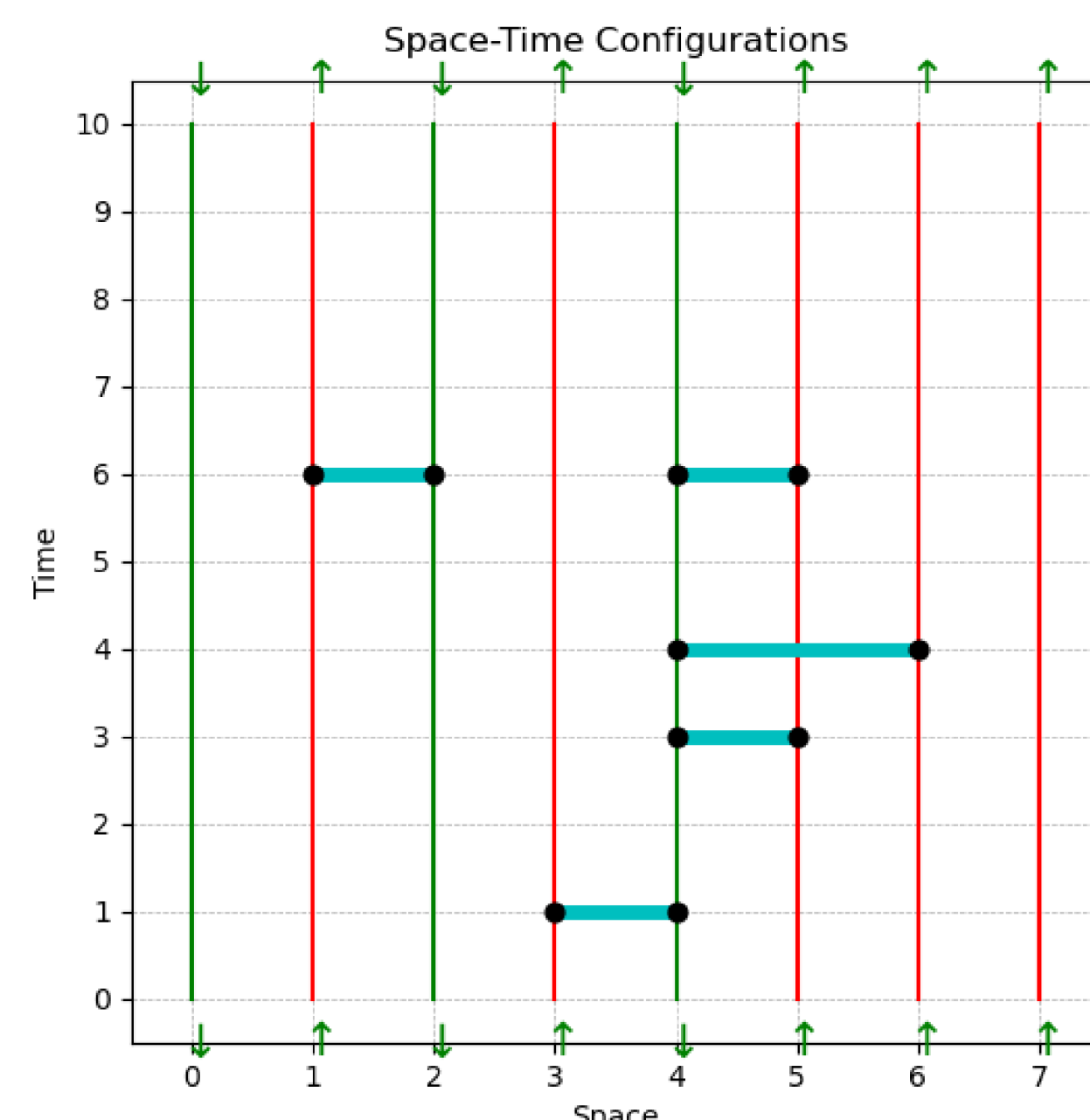


Figure: Sequence of plots visualizing the loop update mechanism.

Results, Analysis, and Discussions

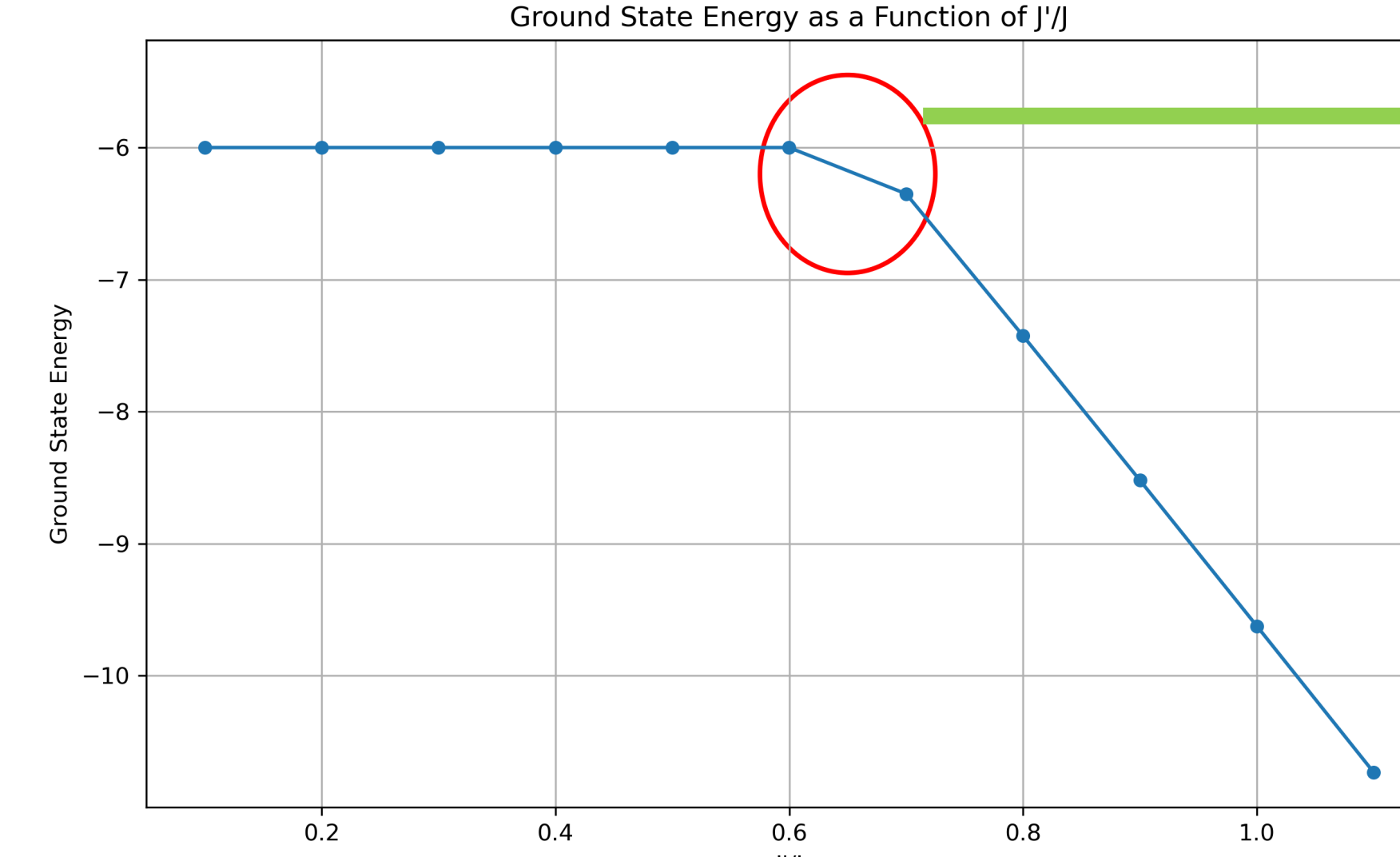


Figure: Ground state energy as a function of the coupling ratio J'/J near the transition.

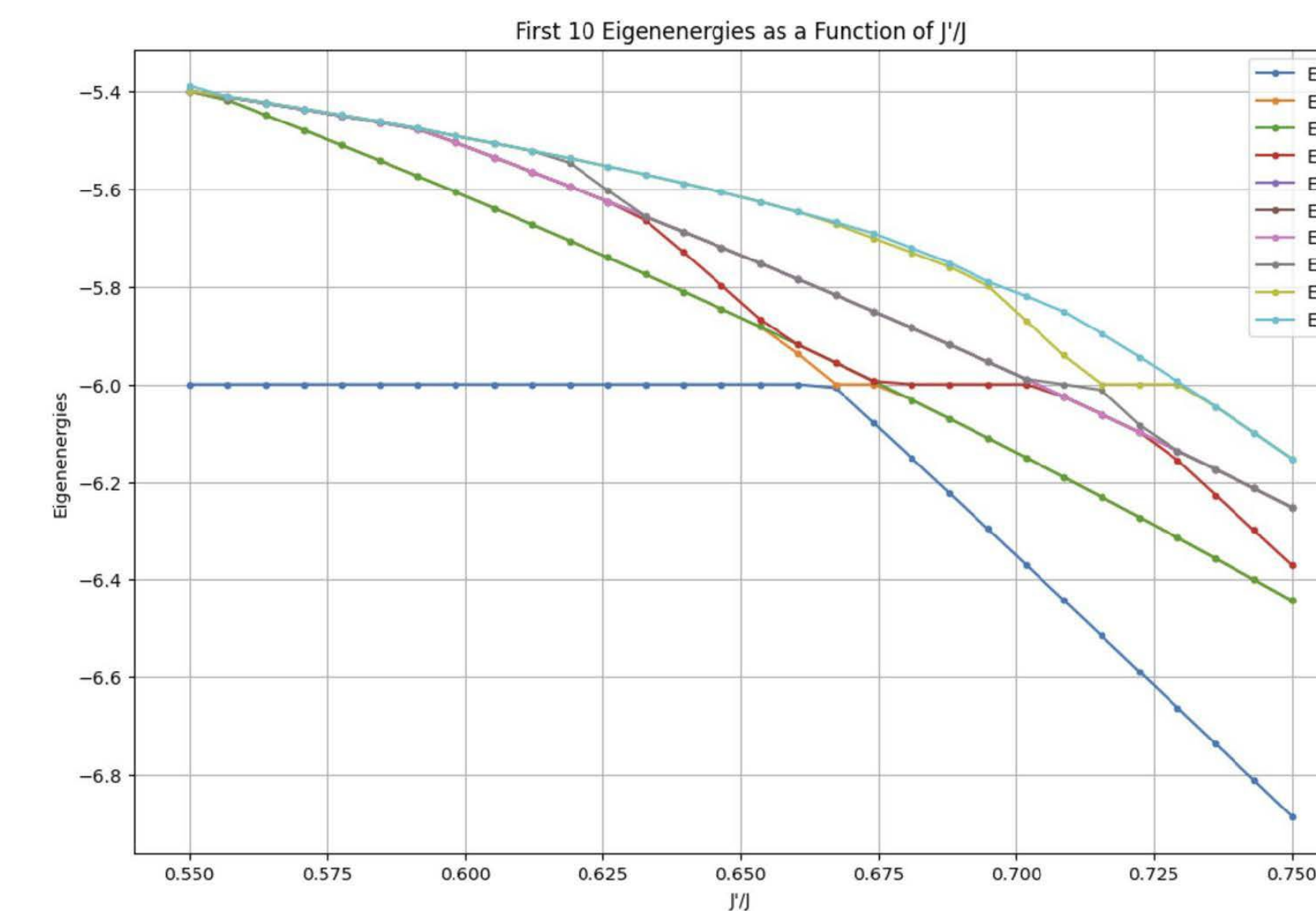


Figure: Lowest 10 eigenenergies of the Shastry-Sutherland model as a function of J'/J .

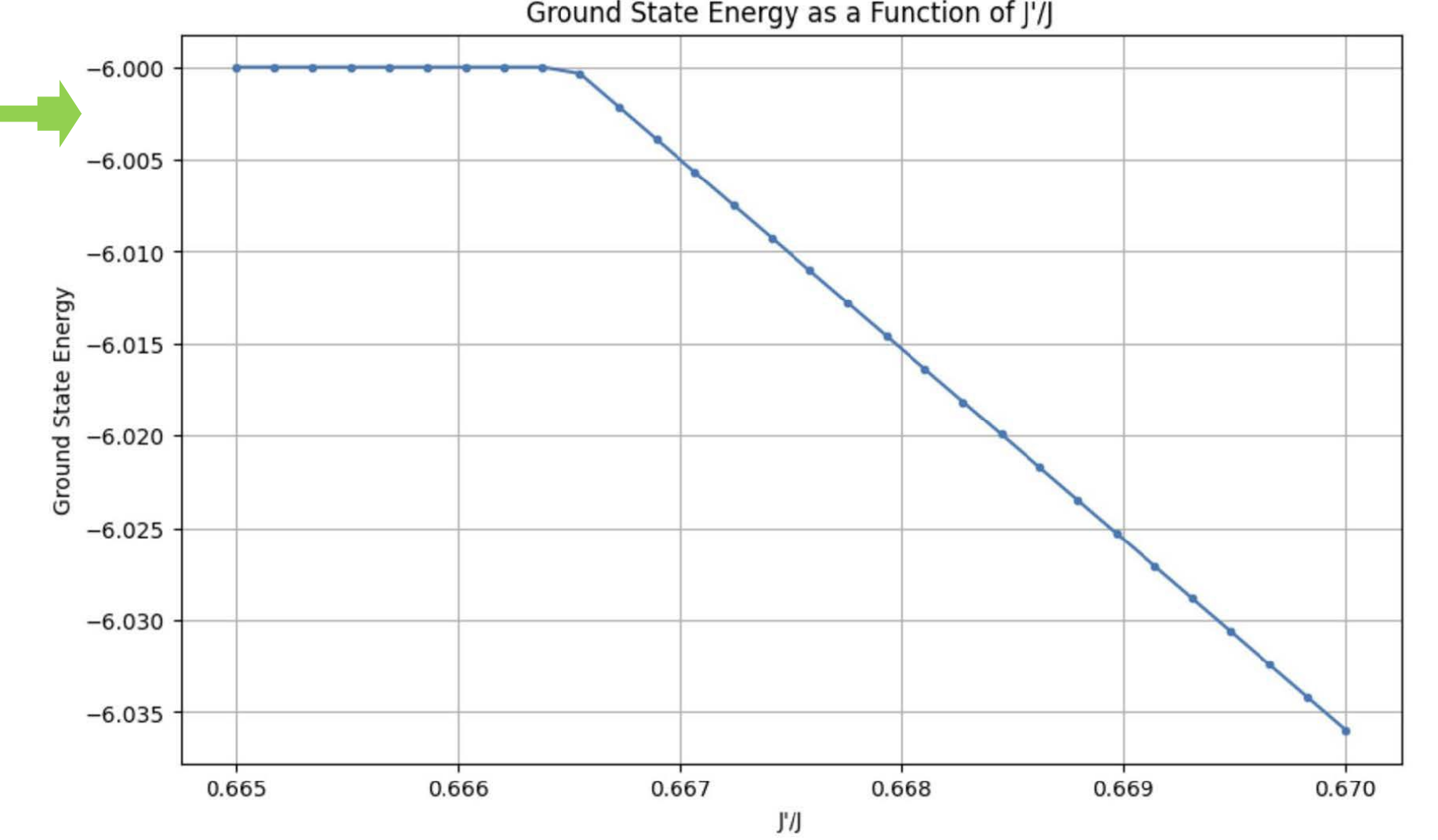


Figure: Ground state energy as a function of the coupling ratio J'/J very near the transition.

Eigenvalue Index	Eigenvalue	Spin s
At $J'/J = 0.5$		
0	-6.0000	0
1	-5.3203	1
2	-5.3200	1
3	-5.1826	0
4	-5.1173	1
5	-5.0296	0
6	-5.0270	0
7	-5.0261	0
At $J'/J = 0.66$		
0	-6.0000	0
1	-5.9331	0
2	-5.9170	0
3	-5.7828	0
4	-5.6456	1
5	-5.6423	0
6	-5.6394	1

Table: Eigenvalues and corresponding total spin s values near the level crossing.

Results and Discussions

- An ED algorithm was implemented on a 4x4 SSM lattice and its energy spectrum was obtained.
- The results matched the theoretical calculations in the analytic regime and energy level crossing diagrams were plotted in the non-analytic regime.
- Some simple yet interesting properties of the Hamiltonian were uncovered, including the fact that the dimer product state is an exact eigenstate.
- States in the singlet sector started reorganizing as J'/J increased above the critical point ~ 0.66 .
- A continuous-time QMC algorithm following the worldline approach has been developed.
- The code to implement the QMC program on larger lattices is under development.

Conclusions

- Exact Diagonalization (ED) algorithms can help solve a wide variety of Quantum Many-body Problems on small system sizes.
- By implementing ED on a 4x4 Shastry-Sutherland Lattice, we obtained the exact ground state and probed the challenging regime.
- The dimer-product state was verified analytically and numerically to be an exact eigenstate for all values of J'/J with eigenvalue $-\frac{3}{8}N_s$.
- Around $J'/J \approx 0.6 - 0.7$, the dimer-product state is replaced by another singlet state to be the ground state.
- A QMC method to explore larger lattice sizes is under rapid development.

References

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